

# Classification Algorithm: Tutor-Feedback v2

Transparent eight-category, multi-label baseline | Prepared collaboratively by the eight-member CS-44 project team

**Team delivery: eight members held defined primary workstreams and completed taxonomy decisions, evidence review, quality assurance, and final acceptance collectively. Core decision: categories are not mutually exclusive. Simulation and Case-Based Learning are separate. Week 4 remains an INFS6600-only rerun.**

## 1. Tutor-directed changes

- Split Simulation from Case-Based Learning.
- Allow a unit and an evidence item to belong to more than one category.
- Expect INFS6600 in Work-Integrated and Applied, Project- and Problem-Based, and Case-Based Learning.
- Add category-level total scoring.
- Configure the remaining supplied taxonomy categories without analysing additional units.

### Supporting and suggested pilot adjustments

- Count distinct evidence items rather than raw phrase occurrences.
- Retain overlapping industry/case evidence instead of forcing a single label.

## 2. Evidence unit and counting policy

One distinct outline item (overview paragraph, learning outcome, assessment row, or weekly-schedule row) evaluated once per category.

Each outline item is evaluated once per category. A rule contributes its weight once even when several alternative phrases are present. Deduplication is at item-category level, so the same item can count once in several categories.

## 3. Taxonomy v2

| Category                                    | Definition                                                                                                                                                                          | Overlap / review rule                                                                                                                                                                                                                                                    |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Work-Integrated and Applied Learning</b> | An educational approach that explicitly merges theory with real-world practice and embeds authentic industry, workplace, or community-relevant work and tasks into a unit of study. | May overlap with Project- and Problem-Based Learning, Case-Based Learning, Community Learning, and Entrepreneurial Learning. Check that professional or partner language describes an authentic learning task rather than a generic context statement.                   |
| <b>Simulation</b>                           | An instructional method that recreates or mimics real-world scenarios, environments, or processes so learners can practise skills and make decisions safely.                        | Separated from Case-Based Learning by tutor revision. A real-world scenario alone is not a simulation unless the activity recreates, mimics, or role-plays a setting or process. Do not inherit Simulation from a case-study label or from Case-Based Learning evidence. |

| Category                                   | Definition                                                                                                                                               | Overlap / review rule                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Case-Based Learning</b>                 | An active teaching method in which students use real-world scenarios and case studies to build critical-thinking and problem-solving skills.             | May overlap with Work-Integrated and Applied Learning and Project- and Problem-Based Learning. Industry context does not exclude a case-based classification. An administrative assessment type such as 'Case studies' is retained as a weak review signal and is not sufficient by itself for a positive classification.             |
| <b>Project- and Problem-Based Learning</b> | Learning focused on real-world problems and challenges using problem-solving, decision-making, investigative, and self-directed learning skills.         | May overlap with Work-Integrated and Applied Learning when projects involve clients or industry partners, and with Case-Based Learning when cases are used inside a project. Do not classify from the generic word 'project' alone; require a problem, challenge, immersion, investigation, or explicit project/problem-based method. |
| <b>Community Learning</b>                  | Learning undertaken together with or for communities, including service-learning, field studies, and professional communities of practice.               | May overlap with Work-Integrated and Applied Learning when the authentic partner is a community organisation. Require an explicit community, service-learning, field-study, or community-of-practice activity.                                                                                                                        |
| <b>Entrepreneurial Learning</b>            | Learning that develops the knowledge, skills, and mindset to recognise opportunities, turn ideas into action, and manage new ventures under uncertainty. | May overlap with Project- and Problem-Based Learning and Simulation when learners develop or simulate a venture. Generic ideation, pitching, or prototyping is a review signal unless explicitly connected to entrepreneurship, a startup, a product-development project, or a venture.                                               |
| <b>Technology-Mediated Learning</b>        | An instructional environment in which digital devices, networks, or specialised technologies mediate access to content and interaction during learning.  | May overlap with Simulation, Hybrid Learning, or Entrepreneurial Learning. Merely studying digital technology is not evidence that learning is technology-mediated. Require the technology to be part of the learning method, not only the subject matter.                                                                            |
| <b>Hybrid Learning</b>                     | An education model that mixes face-to-face classroom teaching with online digital instruction.                                                           | May overlap with Technology-Mediated Learning, but requires an explicit combination of physical and online modes. Online materials alone are insufficient; require an explicit hybrid, blended, flipped, hyflex, or mixed-mode design.                                                                                                |

## 4. Tutor-adjusted WIL weights

| Rule group                                  | v2 weight | Status                |
|---------------------------------------------|-----------|-----------------------|
| industry or business partner                | 4.0       | existing_high_weight  |
| actual organisation or professional context | 4.0       | existing_high_weight  |
| authentic practice                          | 4.0       | tutor_comment_give_4  |
| professional presentation                   | 3.5       | unchanged             |
| theory-practice integration                 | 2.0       | tutor_comment_give_2  |
| practical teamwork                          | 2.0       | tutor_comment_give_2  |
| career readiness                            | 2.0       | tutor_comment_maybe_2 |
| project immersion                           | 2.0       | unchanged             |

## 5. Decision thresholds and confidence

**Positive  $\geq 3.0$ ; high confidence  $\geq 5.0$ . Status: provisional\_pending\_client\_confirmation.**  
**These values remain configurable and are not represented as client-validated cut-offs.**

## 6. Scoring and aggregation

- Normalise lowercase, whitespace, apostrophes, and dash variants.
- Sum weights for matched rule groups within one item-category decision.
- Send an item to review when it is below the positive threshold or contains only review-required rules.
- Classified score total = sum of positive item scores for the unit-category pair.
- Review score total is shown separately and never silently added to the positive score.

## 7. Case-Based versus Simulation

Open-ended business scenarios support Case-Based Learning. A Simulation result requires simulation-specific evidence such as a recreated environment, simulation activity, or role play. The administrative assessment type 'Case studies' is assigned weight 2.0 and remains in the review queue unless stronger case-method evidence appears in the same item.

## 8. INFS6600 regression checks

| Acceptance check                                    | Result |
|-----------------------------------------------------|--------|
| Eight taxonomy categories loaded                    | PASS   |
| Expected three INFS6600 categories present          | PASS   |
| Simulation evaluated independently and not positive | PASS   |
| Administrative case labels remain review-only       | PASS   |
| Thresholds labelled provisional                     | PASS   |

## 9. Open client decisions

- Confirm whether the 3.0 positive and 5.0 high-confidence thresholds should remain.
- Confirm the tentative Career readiness weight of 2.0.
- Confirm whether administrative 'Case studies' labels should remain at review weight 2.0.

## 10. Audit trail

Taxonomy version: 2026-09-01-tutor-feedback-v2

Official source: <https://www.sydney.edu.au/units/INFS6600/2026-S2C-NE-CC>

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